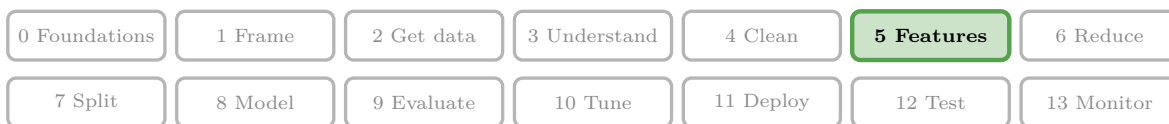


Column Transformer

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Where this fits: Pipeline map step 5 (Engineer features). See the [Course map](#).



- **Builds on:** Simple imputation (mean, median, mode, constant) ([Note 23](#)); Binning and binarization ([Note 23](#)); Encoding categorical data ([Note 26](#)).
- **Leads to:** ML pipelines ([Note 29](#)).

1. Overview

Key point: A column transformer sends each column to its own transformer and joins all the outputs into one array, in a single step.

Real datasets mix feature types. A **feature** is an input variable, one column of the data table. One feature has missing values, another is ordinal, a third is nominal, and a fourth needs nothing at all. Each one needs a different transformation.

Figure 1 shows the whole topic. Every column goes to the transformer that fixes its problem, and scikit-learn's **ColumnTransformer** runs all of them at once and joins their outputs into one array of numbers.

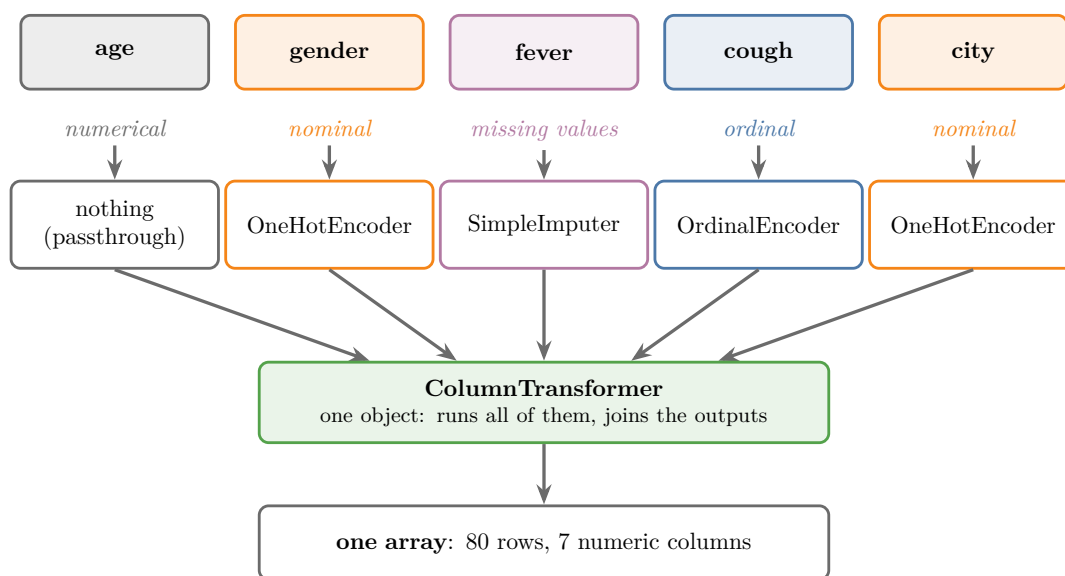


Figure 1: Each column of the COVID data and the transformer it needs; ColumnTransformer runs them all and joins the outputs

2. Different columns need different transformations

Key point: When columns need different transformations, doing them one by one means many separate arrays that we must glue back together.

Take customer data with four features:

- **age:** numerical, but some values are missing. Age needs simple imputation to fill them.
- **city:** nominal categories. City needs one-hot encoding.
- **gender:** nominal categories. Gender also needs one-hot encoding.
- **review:** ordinal categories, such as good and excellent. Review needs ordinal encoding.

If we apply each transformation separately, we get three separate NumPy arrays: one from imputing age, one from one-hot encoding city and gender, and one from ordinal encoding review. We then have to join the three into one big array before we can train a model.

Gluing arrays by hand is a lot of manual work, and the work grows with every feature. Think of a kitchen where each cook prepares one ingredient: a column transformer is the head chef who hands each ingredient to the right cook and plates the results together. A **column transformer** is a scikit-learn class that does the whole job in one go: it takes the DataFrame and returns one finished array.

3. The COVID toy data

Key point: 100 made-up patients; fever has missing values, cough is ordinal, gender and city are nominal, and age is ready as it is.

The data is a small made-up dataset of 100 patients. Each patient is one **observation** (one record, one row of the table). There are six columns: the features age, gender, fever (body temperature in degrees Fahrenheit at the doctor's visit), cough and city, and the **target** `has_covid` (Yes or No), the output we would predict. The dataset is a toy for practising the technique, not real medical data.

age	gender	fever	cough	city	has_covid
60	Male	103.0	Mild	Kolkata	No
27	Male	100.0	Mild	Delhi	Yes
42	Male	101.0	Mild	Delhi	No

A few quick counts describe the columns:

- **cough:** 62 patients have a Mild cough and 38 a Strong one.
- **city:** four cities: Kolkata (32), Bangalore (30), Delhi (22) and Mumbai (16).
- **fever:** 10 of the 100 values are missing.

Python: Loading the data and counting values.

```
import pandas as pd
```

```
df = pd.read_csv("data/covid_toy.csv")
# how many rows hold each category
df["cough"].value_counts()
df["city"].value_counts()
# missing values per column: fever has 10
df.isnull().sum()
```

3.1 What each column needs

Key point: Four features need a transformation; age needs none.

Column	Type	Problem	Transformation
age	numerical	none	none
fever	numerical	10 missing values	simple imputation (SimpleImputer)
cough	ordinal: Mild < Strong	text	ordinal encoding (OrdinalEncoder)
gender	nominal	text	one-hot encoding (OneHotEncoder)
city	nominal	text	one-hot encoding (OneHotEncoder)

The target `has_covid` is categorical too, so it would need label encoding. We leave it out here: the focus is the features.

3.2 Split before transforming

Key point: As always, we split first; every transformer learns from the training set only.

We split the data into a training set (80 rows) and a test set (20 rows) before any transformation. The features are every column except `has_covid`.

Python: Splitting the data.

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    df.drop(columns=["has_covid"]), # inputs
    df["has_covid"],               # target
    test_size=0.2, random_state=0)
# X_train: (80, 5), X_test: (20, 5)
```

4. The hard way: one transformer at a time

Key point: Without a column transformer, we transform each column separately, get one array per step, and glue the arrays together with `np.concatenate`.

Figure 2 shows the plan. Four separate steps each give an array of 80 rows, and a fifth step joins them into one array of 80 rows and 7 columns.

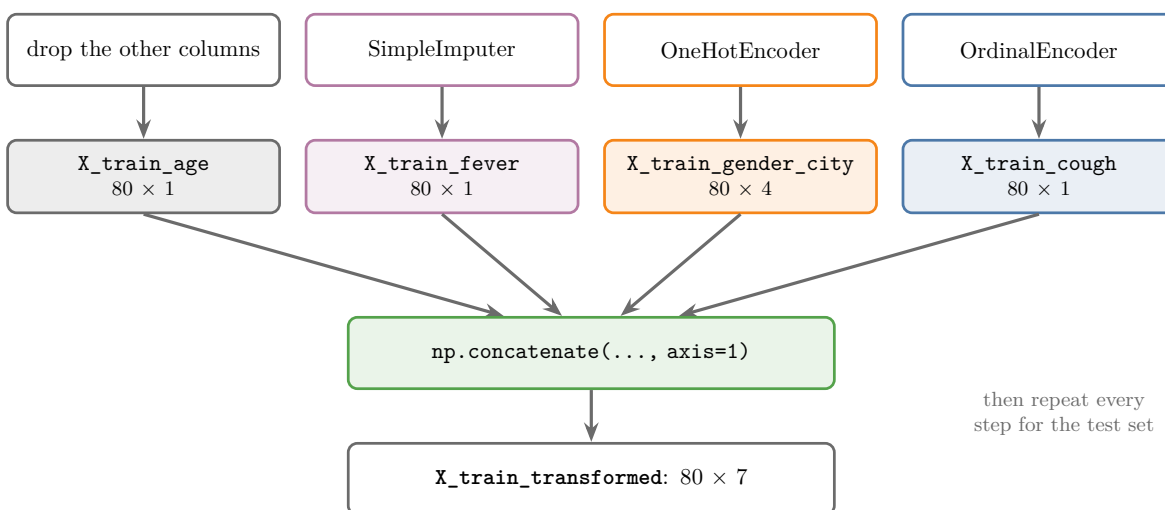


Figure 2: Transforming by hand: one array per step, joined at the end

4.1 Fever: fill the missing values

Key point: `SimpleImputer` replaces each missing fever with the mean fever of the training set.

`SimpleImputer` (with its default setting) fills every missing value with the column's mean. We pick only the `fever` column, fit the imputer on the training set and transform both sets.

Python: Imputing fever.

```
from sklearn.impute import SimpleImputer

si = SimpleImputer()
# double brackets keep a table of one column
X_train_fever = si.fit_transform(X_train[["fever"]])
X_test_fever = si.transform(X_test[["fever"]])
# X_train_fever: (80, 1), no missing values left
```

`fit_transform` does fit and transform in one call. We use it on the training set only; the test set gets just `transform`.

4.2 Cough: ordinal encoding

Key point: Mild becomes 0 and Strong becomes 1.

Cough has an order, Mild < Strong, so we use `OrdinalEncoder` and give it that order.

Python: Ordinal encoding cough.

```
from sklearn.preprocessing import OrdinalEncoder

oe = OrdinalEncoder(categories=[["Mild", "Strong"]])
X_train_cough = oe.fit_transform(X_train[["cough"]])
X_test_cough = oe.transform(X_test[["cough"]])
# X_train_cough: (80, 1)
```

4.3 Gender and city: one-hot encoding

Key point: Gender (2 categories) and city (4 categories) become $1 + 3 = 4$ columns once the first category of each is dropped.

Both columns are nominal, so one `OneHotEncoder` handles the two together. With `drop="first"` it removes one column per feature to avoid multicollinearity.

The column count works out as follows:

- **gender:** 2 categories, minus the first, gives 1 column.
- **city:** 4 categories, minus the first, gives 3 columns.
- **Total:** $1 + 3 = 4$ columns.

Python: One-hot encoding gender and city.

```
from sklearn.preprocessing import OneHotEncoder

ohe = OneHotEncoder(drop="first", sparse_output=False)
X_train_gender_city = ohe.fit_transform(
    X_train[["gender", "city"]])
X_test_gender_city = ohe.transform(
    X_test[["gender", "city"]])
# X_train_gender_city: (80, 4)
sparse_output=False returns an ordinary array instead of a sparse matrix.
```

4.4 Age and the final join

Key point: Age is taken out unchanged, then `np.concatenate` joins the four arrays side by side.

Age needs no transformation, but it still has to become an array so it can be joined to the others. Then `np.concatenate` with `axis=1` puts the arrays next to each other, column after column: $1 + 1 + 4 + 1 = 7$ columns.

Python: Extracting age and joining everything.

```
import numpy as np

X_train_age = X_train[["age"]].values
X_test_age = X_test[["age"]].values

# axis=1: join side by side (column-wise)
X_train_transformed = np.concatenate(
    (X_train_age, X_train_fever,
     X_train_gender_city, X_train_cough), axis=1)
X_test_transformed = np.concatenate(
    (X_test_age, X_test_fever,
     X_test_gender_city, X_test_cough), axis=1)
# (80, 7) and (20, 7)
```

The hand-made way works, but it is hectic for only four features. With 50 columns, it would take a very long time, and every step has to be written twice: once for the training set and once for the test set.

5. The easy way: ColumnTransformer

Key point: We list every transformation once, as (name, transformer, columns), and ColumnTransformer does all the steps of Section 4 in one call.

ColumnTransformer lives in `sklearn.compose`. When we create it, we pass two things: the list of transformers and what to do with the remaining columns.

5.1 The transformers list

Key point: Each transformation is a tuple of three items: a name, a transformer object, and a list of column names.

The parameter **transformers** is a list of tuples, one tuple per transformation. Figure 3 shows the three items inside one tuple.

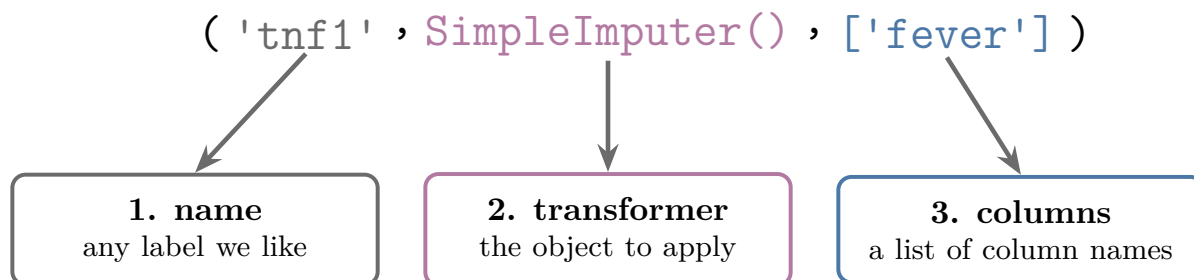


Figure 3: The three items of one tuple in the transformers list

1. **Name:** any label we choose, such as "tnf1".
2. **Transformer:** the object to apply, such as `SimpleImputer()`.
3. **Columns:** a list of the columns to apply it to, such as `["fever"]`.

Our data needs three tuples: one for the imputer, one for the ordinal encoder, and one for the one-hot encoder.

5.2 The remainder parameter

Key point: `remainder` decides what happens to columns that no transformer touches: "drop" removes them, "passthrough" keeps them unchanged.

Sometimes we do not transform every column. The parameter `remainder` says what to do with the rest:

- **"drop":** remove them from the output. Dropping is the default.
- **"passthrough":** keep them as they are.

Here age has no transformer. We want to keep it, so we use `remainder="passthrough"`. With "drop", age would disappear and the output would have 6 columns.

5.3 Building and using the column transformer

Key point: `fit_transform` on the training set and `transform` on the test set give the same 80×7 and 20×7 arrays as the hard way.

Python: The whole of Section 4 in one object.

```
from sklearn.compose import ColumnTransformer

transformer = ColumnTransformer(transformers=[
    ("tnf1", SimpleImputer(), ["fever"]),
    ("tnf2", OrdinalEncoder(
        categories=[["Mild", "Strong"]]), ["cough"]),
    ("tnf3", OneHotEncoder(
        sparse_output=False, drop="first"),
        ["gender", "city"]),
], remainder="passthrough")

# learn from the training set and transform it
transformer.fit_transform(X_train).shape    # (80, 7)
# apply to the test set
transformer.transform(X_test).shape         # (20, 7)
```

Figure 4 shows what happens inside, using three training rows, one of them with a missing fever. The columns are routed to their transformers, each transformer works only on its own columns, and the outputs are stitched side by side. That missing fever is filled with 100.9, the mean fever of the training set.

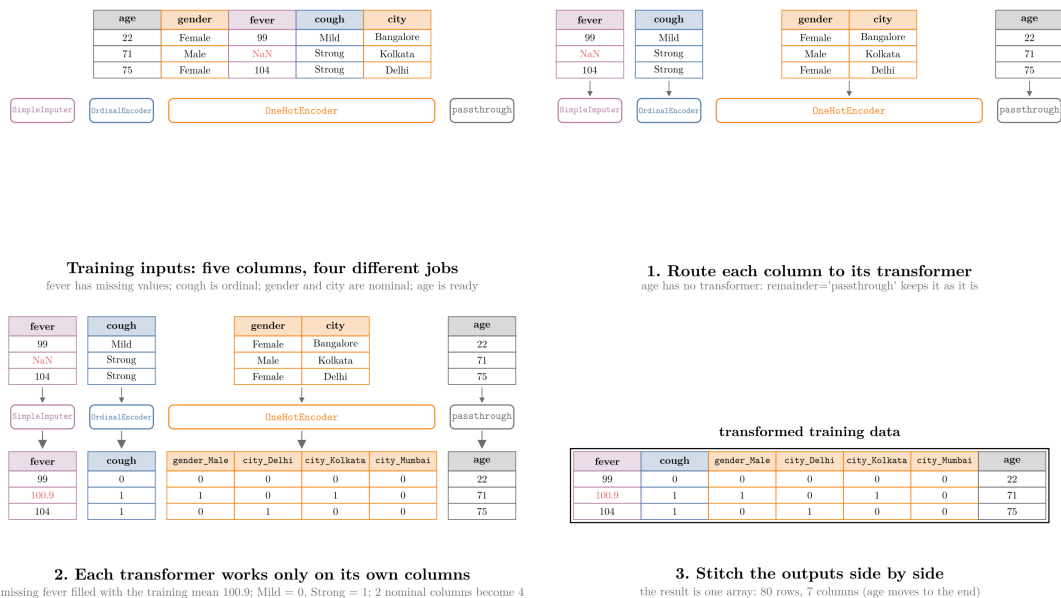


Figure 4: ColumnTransformer step by step: route, transform, stitch

On the real output, the first three training patients look like this before the transformation:

Patient	age	gender	fever	cough	city
1	22	Female	99.0	Mild	Bangalore
2	56	Female	104.0	Strong	Bangalore
3	31	Female	98.0	Mild	Kolkata

After it, they are three rows of 7 numbers. The table below is turned on its side to fit: each line is one output column, and each number column is one patient.

Output column	Patient 1	Patient 2	Patient 3
fever	99	104	98
cough	0	1	0
gender_Male	0	0	0
city_Delhi	0	0	0
city_Kolkata	0	0	1
city_Mumbai	0	0	0
age	22	56	31

Bangalore is the first city alphabetically, so it was the dropped category: a Bangalore row has 0 in all three city columns.

The column transformer combines naturally with another scikit-learn tool, the pipeline, covered in the next Note. Used together, they make preprocessing much simpler.

Extra: The output columns come in the order of the `transformers` list, and the `remainder` columns go last. So the column transformer puts age at the end, while our hand-made array had it first. Both arrays hold the same 7 columns, only in a different order.

6. Reading the output

Key point: `get_feature_names_out` gives the name of every output column, and `set_output(transform="pandas")` returns a DataFrame with those names instead of a bare array.

Extra: Section 6 goes beyond the basics: it uses newer scikit-learn tools that make the output easier to read.

The result of `fit_transform` is a NumPy array without column names, so it is easy to forget which column is which. Two tools fix this.

6.1 Names of the output columns

Key point: Each output name is the transformer's name, two underscores, then the column name.

`get_feature_names_out` lists the output columns in order. Each name starts with the transformer's name and two underscores; columns kept by `remainder` start with `remainder__`.

Python: Listing the output column names.

```
transformer.get_feature_names_out()
# ['tnf1__fever', 'tnf2__cough',
#  'tnf3__gender_Male', 'tnf3__city_Delhi',
#  'tnf3__city_Kolkata', 'tnf3__city_Mumbai',
#  'remainder__age']
```

6.2 A DataFrame instead of an array

Key point: After `set_output(transform="pandas")`, the column transformer returns a DataFrame.

Calling `set_output(transform="pandas")` once makes every later `fit_transform` and `transform` return a pandas DataFrame, with the names above as its columns.

Python: Getting a DataFrame back.

```
transformer.set_output(transform="pandas")
transformer.fit_transform(X_train).head()
# a DataFrame with columns tnf1__fever ... remainder__age
```

Passing `verbose_feature_names_out=False` to `ColumnTransformer` drops the prefixes, giving plain names such as `fever` and `city_Delhi`.

Extra: The fitted transformers are kept inside the column transformer, under their names. For example, `transformer.named_transformers_["tnf1"].statistics_` gives `[100.92]`, the training mean used to fill the missing fevers.

7. Summary

	One transformer at a time	ColumnTransformer
Objects	one per transformation	one
Choosing columns	by hand, for each step	in the <code>transformers</code> list
Untouched columns	extracted by hand	<code>remainder="passthrough"</code>
Joining the outputs	<code>np.concatenate</code> by hand	automatic
Test set	every step written again	one <code>transform</code> call
Result for this data	80×7 (age first)	80×7 (age last)

- Different columns often need different transformations: imputation, ordinal encoding, one-hot encoding, or none.
- Done by hand, each transformation gives its own array, and we must join them with `np.concatenate`, for both sets.
- `ColumnTransformer` takes a list of (name, transformer, columns) tuples and does every step in one call.
- `remainder="drop"` (default) removes untouched columns; `remainder="passthrough"` keeps them, at the end.
- Split first; `fit_transform` on the training set, `transform` on the test set.
- `get_feature_names_out` and `set_output(transform="pandas")` tell us which output column is which.

8. Key terms

Term	Meaning
Feature	An input variable, one column of the data table
Target	The output we predict
Observation	One record, one row of the data table
Column transformer	A scikit-learn class that applies a different transformer to each set of columns and joins the outputs into one array
ColumnTransformer	The scikit-learn class (in <code>sklearn.compose</code>) that implements the column transformer
transformers	The <code>ColumnTransformer</code> parameter: a list of (name, transformer, columns) tuples
remainder	The <code>ColumnTransformer</code> parameter for untouched columns: <code>"drop"</code> (default) or <code>"passthrough"</code>
passthrough	The <code>remainder</code> option that keeps untouched columns unchanged
SimpleImputer	scikit-learn's class that fills missing values, by default with the column's mean

Term	Meaning
<code>fit_transform</code>	Fit and transform in one call; used on the training set only
<code>np.concatenate</code>	NumPy function that joins arrays; with <code>axis=1</code> it puts them side by side
<code>get_feature_names_out</code>	Method returning the names of the output columns, in order
<code>set_output</code>	Method that makes a transformer return a pandas DataFrame with <code>transform="pandas"</code>